

Exp No: 1**Study of various Network commands used****AIM:**

To study and practise the various network commands used in Linux and Windows operating systems.

ALGORITHM / PROCEDURE:

1. Open the Terminal in Linux and the Command Prompt (cmd) in Windows.
2. Use hostname to see the name of the computer.
3. Use ifconfig (Linux) and ipconfig (Windows) to see the IP address, subnet mask and MAC address.
4. Use ping to check whether the remote system is reachable. Note the packet loss and the time taken.
5. Use traceroute (Linux) and tracert (Windows) to see the path taken by the packets.
6. Use nslookup and dig to find the IP address of a domain name.
7. Use netstat to see the active connections and open ports.
8. Use arp -a to see the IP address to MAC address table.
9. Use route (Linux) and route print (Windows) to see the routing table.
10. Note down the output of each command.

Commands used:

Purpose	Linux	Windows
Host name	hostname	hostname
IP configuration	ifconfig / ip addr	ipconfig /all
Check connectivity	ping -c 4 google.com	ping google.com
Trace the route	traceroute google.com	tracert google.com
DNS lookup	nslookup / dig	nslookup
Active connections	netstat -tuln	netstat -an
ARP table	arp -a	arp -a
Routing table	route -n	route print
MAC address	ip link	getmac

DIAGRAM:

Fig 1: Network commands and the layers they work in

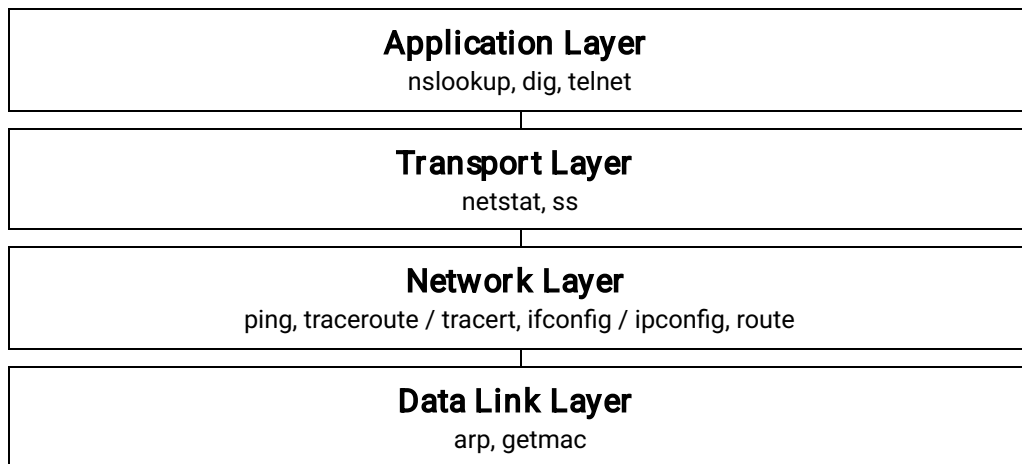
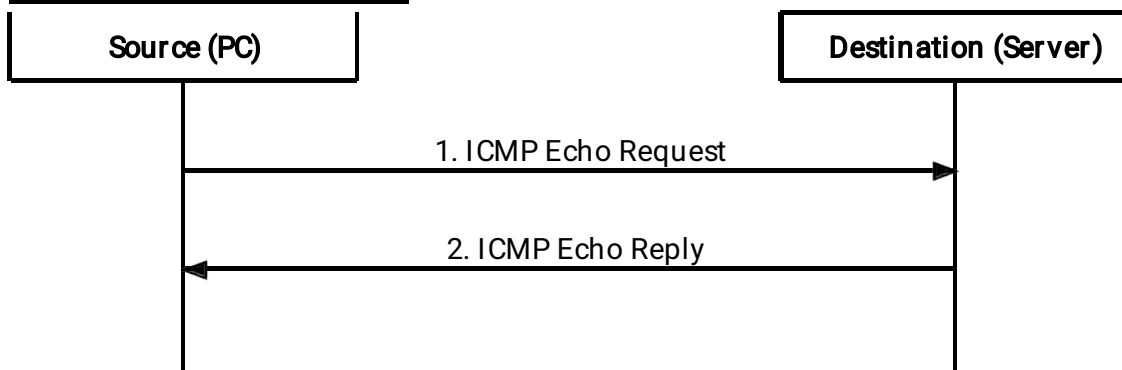


Fig 2: Working of ping command



OUTPUT:

Linux:

```
USER@USER-VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ hostname
USER -VirtualBox
```

```
USER@USER-VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.10 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fe4e:66a1 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:4e:66:a1 txqueuelen 1000 (Ethernet)
    RX packets 2543 bytes 2154321 (2.1 MB)
    TX packets 1987 bytes 312456 (312.4 KB)
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
```

```
USER @USER -VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ ping -c 4 google.com
PING google.com (142.250.195.78) 56(84) bytes of data.
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=1 ttl=117 time=14.2 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=2 ttl=117 time=14.8 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=3 ttl=117 time=13.9 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=4 ttl=117 time=15.2 ms

-- google.com ping statistics --
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 13.904/14.525/15.230/0.512 ms
```

```
USER@USER-VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ traceroute google.com
traceroute to google.com (142.250.195.78), 30 hops max, 60 byte packets
 1 _gateway (192.168.1.1) 1.102 ms 1.045 ms 0.987 ms
 2 10.20.0.1 (10.20.0.1) 4.321 ms 4.287 ms 4.410 ms
 3 172.16.5.9 (172.16.5.9) 8.765 ms 8.801 ms 8.730 ms
 4 72.14.215.85 (72.14.215.85) 12.543 ms 12.498 ms 12.610 ms
 5 maa03s45-in-f14.1e100.net (142.250.195.78) 14.230 ms 14.187 ms 14.301 ms
```

```
USER@USER-VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ nslookup google.com
Server: 192.168.1.1
Address: 192.168.1.1#53

Non-authoritative answer:
Name: google.com
Address: 142.250.195.78
```

```
USER@USER-VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ dig google.com +short
142.250.195.78
```

```
USER@USER-VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address Foreign Address State
tcp 0 0 0.0.0.0:22 0.0.0.0:* LISTEN
tcp 0 0 127.0.0.1:631 0.0.0.0:* LISTEN
tcp6 0 0 :::80 :::* LISTEN
udp 0 0 0.0.0.0:68 0.0.0.0:*
```

```
USER@USER-VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ arp -a
_gateway (192.168.1.1) at c4:e9:84:2a:5b:10 [ether] on enp0s3
? (192.168.1.12) at 3c:52:82:7d:11:9e [ether] on enp0s3
```

```
USER@USER-VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
0.0.0.0 192.168.1.1 0.0.0.0 UG 100 0 0 enp0s3
192.168.1.0 0.0.0.0 255.255.255.0 U 100 0 0 enp0s3
```

Windows:

Command Prompt

```
C:\Users\USER >hostname
DESKTOP-USER
```

```
C:\Users\USER >ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : DESKTOP-USER
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
```

Ethernet adapter Ethernet:

```
Physical Address. . . . . : 00-1A-2B-3C-4D-5E
DHCP Enabled. . . . . : Yes
IPv4 Address. . . . . : 192.168.1.25(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DNS Servers . . . . . : 8.8.8.8
```

Command Prompt

```
C:\Users\USER >ping google.com
```

```
Pinging google.com [142.250.195.78] with 32 bytes of data:
Reply from 142.250.195.78: bytes=32 time=15ms TTL=117
Reply from 142.250.195.78: bytes=32 time=14ms TTL=117
Reply from 142.250.195.78: bytes=32 time=14ms TTL=117
Reply from 142.250.195.78: bytes=32 time=16ms TTL=117
```

```
Ping statistics for 142.250.195.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 16ms, Average = 14ms
```

Command Prompt

```
C:\Users\USER >tracert google.com
```

```
Tracing route to google.com [142.250.195.78]
over a maximum of 30 hops:
```

```
 1 <1 ms <1 ms <1 ms 192.168.1.1
 2  4 ms  4 ms  4 ms 10.20.0.1
 3  9 ms  8 ms  9 ms 172.16.5.9
 4 13 ms 12 ms 12 ms 72.14.215.85
 5 14 ms 14 ms 15 ms 142.250.195.78
```

```
Trace complete.
```

Command Prompt

```
C:\Users\USER >nslookup google.com
Server: dns.google
Address: 8.8.8.8
```

```
Non-authoritative answer:
Name: google.com
Address: 142.250.195.78
```

```
Command Prompt
C:\Users\ USER >netstat -an

Active Connections

 Proto Local Address  Foreign Address  State
TCP 0.0.0.0:135    0.0.0.0:0        LISTENING
TCP 0.0.0.0:445    0.0.0.0:0        LISTENING
TCP 192.168.1.25:50234 142.250.195.78:443 ESTABLISHED
UDP 0.0.0.0:5353    *:*
```

```
Command Prompt
C:\Users\USER >arp -a

Interface: 192.168.1.25 --- 0x8
 Internet Address  Physical Address  Type
192.168.1.1      c4-e9-84-2a-5b-10 dynamic
192.168.1.255    ff-ff-ff-ff-ff-ff static
```

```
Command Prompt
C:\Users\USER >route print

IPv4 Route Table
=====
Active Routes:
 Network Destination  Netmask  Gateway  Interface  Metric
0.0.0.0      0.0.0.0  192.168.1.1  192.168.1.25  25
127.0.0.0    255.0.0.0  On-link  127.0.0.1  331
192.168.1.0  255.255.255.0  On-link  192.168.1.25  281
```

```
Command Prompt
C:\Users\USER >getmac

Physical Address  Transport Name
=====
00-1A-2B-3C-4D-5E \Device\Tcpip_{8F3A21B0-5C6D-4E7F-9A1B-2C3D4E5F6A7B}
```

RESULT:

Thus the various network commands used in Linux and Windows were studied and executed successfully.